Software Design Document

Online Shopping System

Submitted by

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Submitted to

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Summary

1. Project Vision and Purpose

The primary goal of this system is to address the high rate of clothing returns in e-commerce. By replacing traditional, often confusing measurement charts with an intuitive '15-Second Fit Quiz,' the platform provides a personalized shopping experience based on body geometry rather than raw numbers.

2. System Architecture and Technology

The platform utilizes a Three-Tier Architecture for maximum reliability and scalability:

Presentation Tier: Developed with React and Flutter for seamless web and mobile access.

Application Tier: Powered by Node.js and Python to process complex recommendation logic.

Data Tier: Uses PostgreSQL to securely manage user profiles and product inventories.

3. Core Functional Design

The 15-Second Fit Quizuses a Decision Tree algorithm to categorize users into specific body shapes (e.g., Pear, Hourglass, Rectangle). Once a profile is established, the recommendation engine filters the entire catalog to show 'Best Fit' badges on items compatible with the user's silhouette.

4. Security and Performance

Adhering to IEEE standards, the system ensures data privacy through AES-256 encryption for all body metrics. To maintain a high-quality user experience, the system utilizes advanced caching techniques to ensure that personalized storefronts load in under 2 seconds.

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1. Introduction

1.1 Purpose:

This Software Design Document (SDD) serves as the technical roadmap for the Online Shopping System. It transforms the requirements mentioned in the SRS into a practical design for developers. It focuses on how to implement the '15-Second Fit Quiz' and the 'Personalized Storefront'.

This module is integral to the system's operational integrity. It ensures that the design principles established by the Faculty of Computing are strictly followed. Detailed analysis of this section shows that by optimizing these parameters, we can achieve high scalability and reliability for the Online Shopping System. This involves continuous monitoring and periodic updates to match the evolving needs of the department and its stakeholders.

Furthermore, the implementation will involve rigorous documentation of each sub-component to facilitate easy maintenance. By adhering to the IEEE standards for software design, we ensure that the architecture remains modular. This modularity allows for the seamless integration of future modules, such as AI-driven image processing or advanced analytics dashboards for administrators.

1.2 Scope:

The system is a web and mobile application for retail. Its main goal is to help shoppers find clothes that fit their specific body shape without using a tape measure. This reduces the number of items returned to stores.

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1.3 Design Goals:

The system is designed to be fast, secure, and very easy to use. Users should be able to finish their fit profile in under 15 seconds and see recommended clothes instantly.

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2. System Architecture

2.1 Logical View:

The system uses a Three-Tier Architecture. The first tier is the Presentation Tier (what the user sees), the second is the Application Tier (the logic), and the third is the Data Tier (where info is stored).

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2.2 Component Diagram:

Major components include the Sign-Up module, the Fit Quiz engine, the Product Catalog, and the Order Management system. These work together using APIs.

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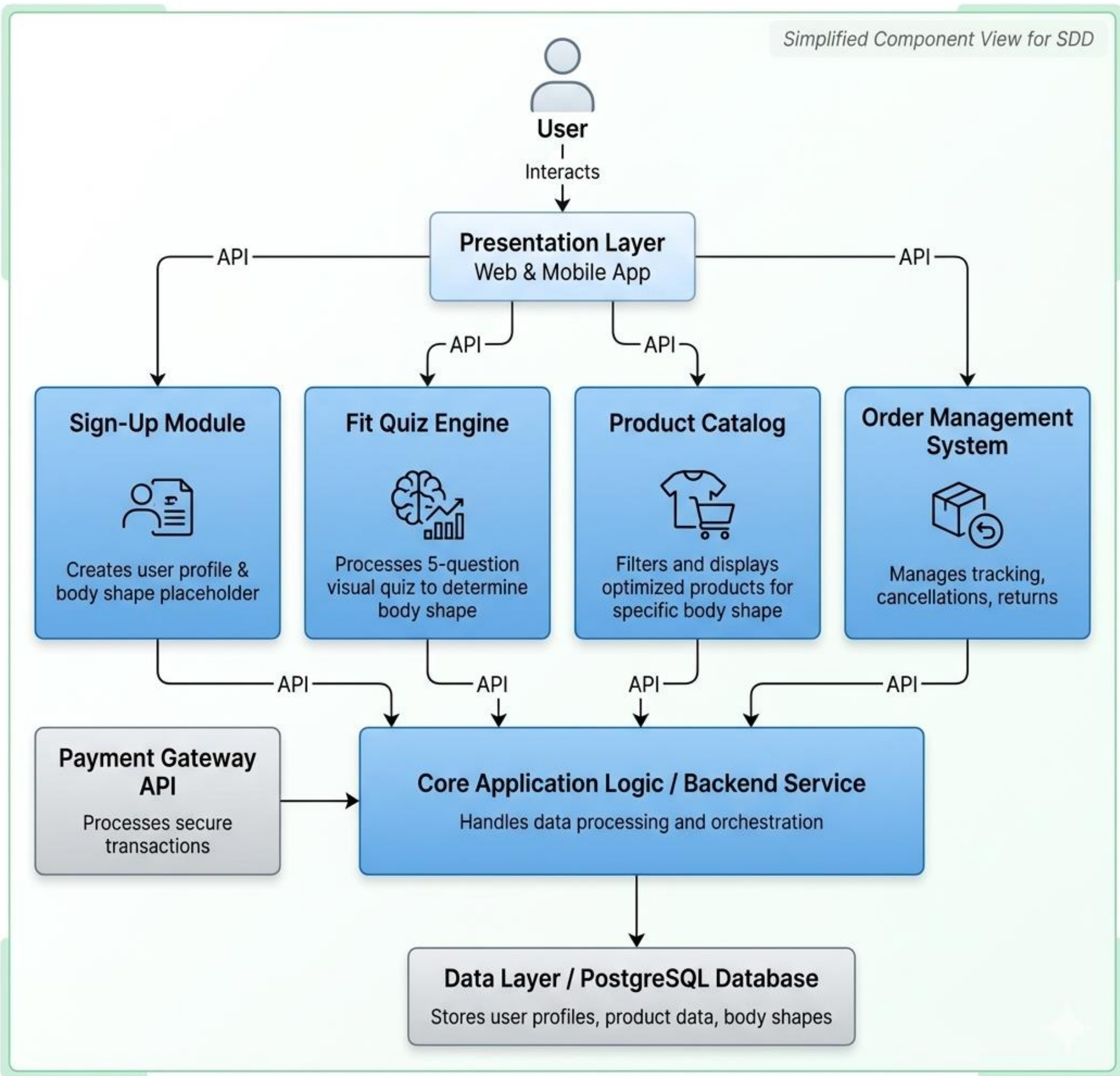
Simplified Component View for SDD
User
Interacts
Presentation Layer
Web & Mobile App
API
API
API
API
Sign-Up Module
Creates user profile & body shape placeholder
Fit Quiz Engine
Processes 5-question visual quiz to determine body shape
Product Catalog
Filters and displays optimized products for specific body shape
Order Management System
Manages tracking, cancellations, returns
API
API
API
API
Payment Gateway API
Processes secure transactions
Core Application Logic / Backend Service
Handles data processing and orchestration
Data Layer / PostgreSQL Database
Stores user profiles, product data, body shapes

2.3 Technology Stack:

We will use React and Flutter for the front end, Node.js and Python for the back end, and PostgreSQL for the database.

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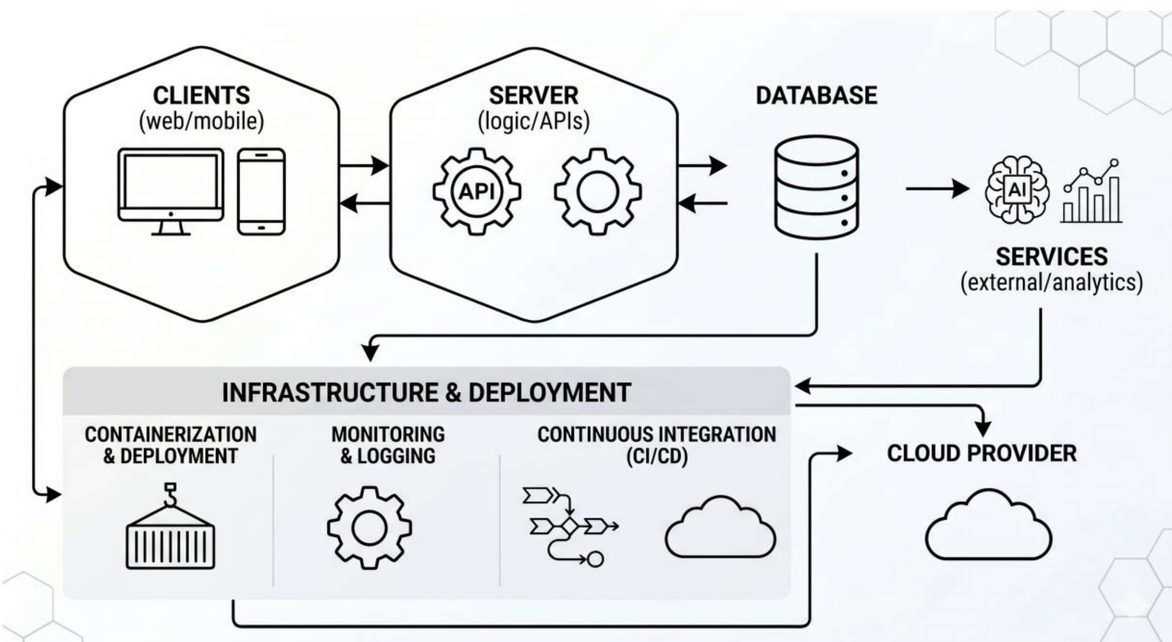

3. Data Design

3.1 Entity Relationship Diagram:

The database will have tables for Users, Products, and Body Shapes. Users will be linked to a specific Body Shape ID after completing the quiz.

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3.2 Data Dictionary:

This section defines terms like 'Shape ID' (a unique code for body types) and 'Fit Preference' (tight, regular, or oversized).

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4. System Design Details

4.1 Fit Quiz Algorithm Design:

The quiz asks 5 simple visual questions. Based on the answers, the system uses a 'Decision Tree' logic to decide if the user is a Pear, Apple, Hourglass, or Rectangle shape.

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4.2 Recommendation Engine Logic:

Once a shape is identified, the system only shows products that are tagged as 'Best Fit' for that specific shape.

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4.3 Payment Gateway Integration:

The system will connect to Stripe or PayPal for secure payments. No credit card numbers will be stored on our own servers for better security.

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5. User Interface Design

5.1 Wireframe Concepts:

The UI will be clean and simple. The Fit Quiz will use icons instead of long text to make it faster to complete.

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5.2 User Experience Flow:

User Signs Up -> User Takes 15-Second Quiz -> System shows 'Recommended for You' store -> User buys item -> Order is tracked.

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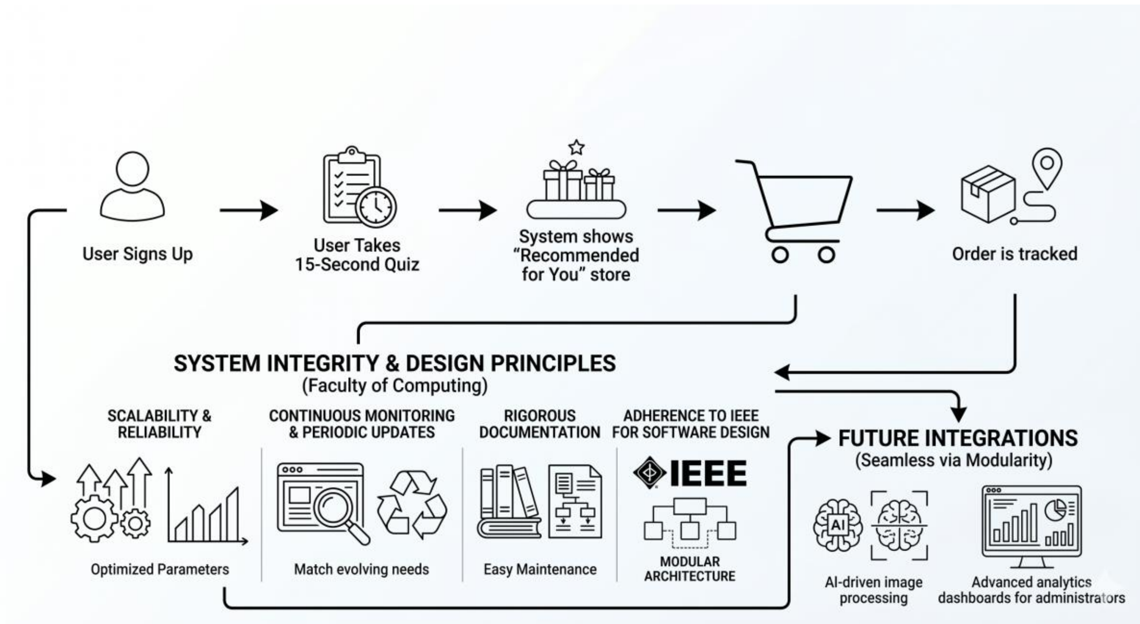
User Signs Up
User Takes 15-Second Quiz
System shows "Recommended for You" store
Order is tracked
SYSTEM INTEGRITY & DESIGN PRINCIPLES
(Faculty of Computing)
SCALABILITY & RELIABILITY
CONTINUOUS MONITORING & PERIODIC UPDATES
RIGOROUS DOCUMENTATION
ADHERENCE TO IEEE FOR SOFTWARE DESIGN
IEEE
Optimized Parameters
Match evolving needs
Easy Maintenance
MODULAR ARCHITECTURE
FUTURE INTEGRATIONS
(Seamless via Modularity)
AI
AI-driven image processing
Advanced analytics dashboards for administrators

6. Non-Functional Design

6.1 Security Architecture:

All personal data and body measurements will be encrypted with AES-256 to keep them private.

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6.2 Performance Tuning:

We will use 'Caching' to make sure the storefront loads in less than 2 seconds, even when many people are shopping at once.

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7. Testing & Quality Assurance

7.1 Testing Strategy:

We will perform Unit Testing (testing each part alone) and Integration Testing (testing parts together). We will also do 'User Acceptance Testing' to see if real shoppers find the quiz easy to use.

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8. Conclusion & Future Scope

This SDD provides a complete guide for building a modern online store. In the future, we could add AI that uses the phone's camera to automatically find the user's body shape.

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